0580 | PAPER 4

TOPIC

04

Geometry

Questions only | Original examination pages retained

29
QUESTIONS

46
EXAM PAGES

3
SECTIONS

USE THE TRACKER | ATTEMPT WITHOUT NOTES | MARK | REVIEW ERRORS

TOPIC CONTENTS

Quick navigation and progress overview

04.1 **Circle geometry** 23 questions | 37 pages

04.2 **Constructions, loci and scale drawings** 4 questions | 6 pages

04.3 **Similarity and congruence** 2 questions | 3 pages

0580 | PAPER 4

SECTION 04.1

Circle geometry

Sorted by session, paper variant and original question number

23
QUESTIONS

37
EXAM PAGES

P4
CALCULATOR

TRACK YOUR ATTEMPTS | WRITE FULL WORKING | CHECK ACCURACY | REVIEW ERRORS

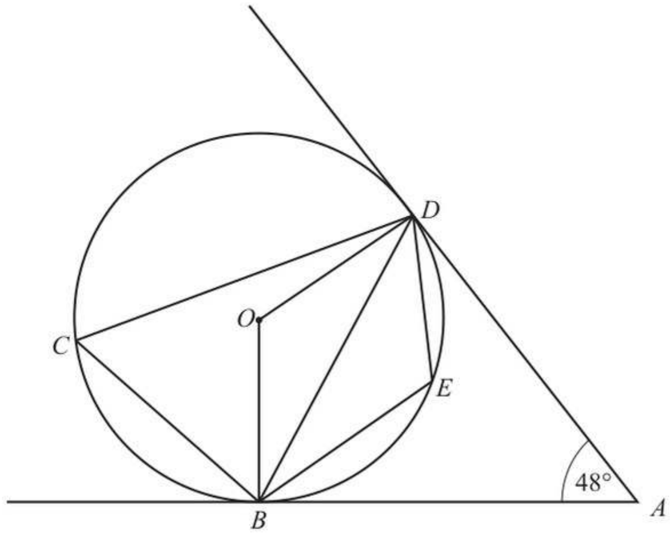
04.1 Circle geometry

#	SOURCE QUESTION	PAGES	DONE	MARK	REVIEW
01	2015 May June Paper 42 | Question 2	1	☐	___	☐
02	2015 May June Paper 43 | Question 6	2	☐	___	☐
03	2015 Oct Nov Paper 41 | Question 5	1	☐	___	☐
04	2015 Oct Nov Paper 43 | Question 8	2	☐	___	☐
05	2016 Oct Nov Paper 42 | Question 8	2	☐	___	☐
06	2017 Oct Nov Paper 41 | Question 2	1	☐	___	☐
07	2018 May June Paper 42 | Question 9	3	☐	___	☐
08	2018 Oct Nov Paper 41 | Question 10	2	☐	___	☐
09	2018 Oct Nov Paper 42 | Question 7	1	☐	___	☐
10	2019 May June Paper 42 | Question 2	1	☐	___	☐
11	2019 May June Paper 42 | Question 7	1	☐	___	☐
12	2019 Oct Nov Paper 41 | Question 1	2	☐	___	☐

04.1 Circle geometry

#	SOURCE QUESTION	PAGES	DONE	MARK	REVIEW
13	2020 May June Paper 43 | Question 8	2	☐	___	☐
14	2020 Oct Nov Paper 42 | Question 8	2	☐	___	☐
15	2020 Oct Nov Paper 43 | Question 5	2	☐	___	☐
16	2021 Feb March Paper 42 | Question 3	2	☐	___	☐
17	2021 Oct Nov Paper 43 | Question 2	1	☐	___	☐
18	2022 Feb March Paper 42 | Question 6	2	☐	___	☐
19	2022 May June Paper 42 | Question 2	1	☐	___	☐
20	2022 Oct Nov Paper 43 | Question 8	2	☐	___	☐
21	2023 Oct Nov Paper 42 | Question 10	1	☐	___	☐
22	2023 Oct Nov Paper 43 | Question 4	1	☐	___	☐
23	2024 Oct Nov Paper 42 | Question 4	2	☐	___	☐

2



NOT TO SCALE

In the diagram, B, C, D and E lie on the circle, centre O.
AB and AD are tangents to the circle.
Angle $BAD = 48°$.

(a) Find

(i) angle ABD,

Answer(a)(i) Angle ABD = ... [1]

(ii) angle OBD,

Answer(a)(ii) Angle OBD = ... [1]

(iii) angle BCD,

Answer(a)(iii) Angle BCD = ... [2]

(iv) angle BED.

Answer(a)(iv) Angle BED = ... [1]

(b) The radius of the circle is 15 cm.

Calculate the area of triangle BOD.

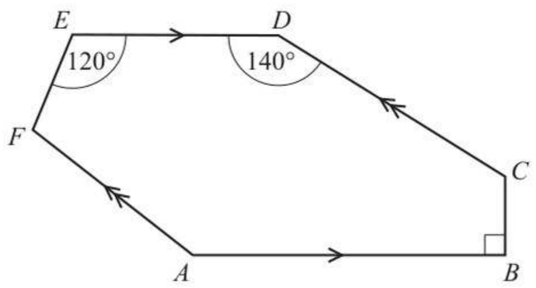
Answer(b) .... cm^2 [2]

(c) Give a reason why $ABOD$ is a cyclic quadrilateral.

Answer(c) ....

.... [1]

6 (a)

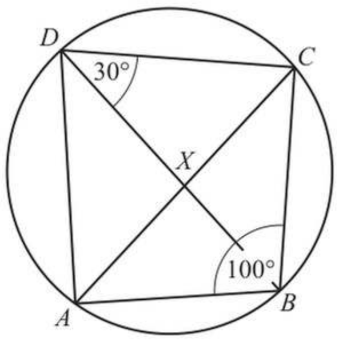


In the hexagon *ABCDEF*, *AB* is parallel to *ED* and *AF* is parallel to *CD*.
Angle *ABC* = 90°, angle *CDE* = 140° and angle *DEF* = 120°.

Calculate angle *EFA*.

Answer(a) Angle *EFA* = ... [4]

(b)



In the cyclic quadrilateral *ABCD*, angle *ABC* = 100° and angle *BDC* = 30°.
The diagonals intersect at *X*.

(i) Calculate angle *ACB*.

Answer(b)(i) Angle *ACB* = ... [2]

(ii) Angle *BXC* = 89°.

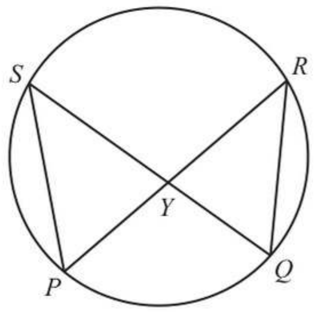
Calculate angle *CAD*.

Answer(b)(ii) Angle *CAD* = ... [2]

(iii) Complete the statement.

Triangles *AXD* and *BXC* are .... [1]

(c)



NOT TO SCALE

P, Q, R and S lie on a circle.
PR and QS intersect at Y.
$PS = 11$ cm, $QR = 10$ cm and the area of triangle $QRY = 23\text{ cm}^2$.

Calculate the area of triangle PYS.

Answer(c) .... cm^2 [2]

(d) A regular polygon has n sides.
Each exterior angle is equal to $\frac{n}{10}$ degrees.

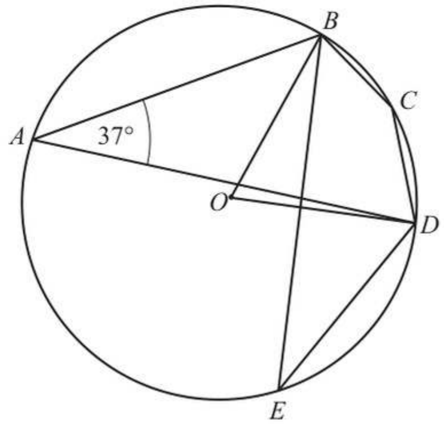
(i) Find the value of n.

Answer(d)(i) $n =$ .... [3]

(ii) Find the size of an interior angle of this polygon.

Answer(d)(ii) .... [2]

5



NOT TO SCALE

A, B, C, D and E are points on the circle, centre O.
Angle $BAD = 37°$.

Complete the following statements.

(a) Angle BED = ... because ....

.... [2]

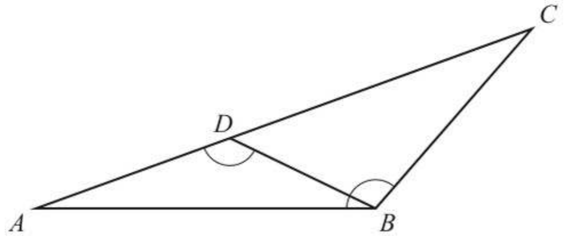
(b) Angle BOD = ... because ....

.... [2]

(c) Angle BCD = ... because ....

.... [2]

8 **(a)**



NOT TO SCALE

In the diagram, D is on AC so that angle ADB = angle ABC.

(i) Show that angle ABD is equal to angle ACB.

Answer(a)(i)

[2]

(ii) Complete the statement.

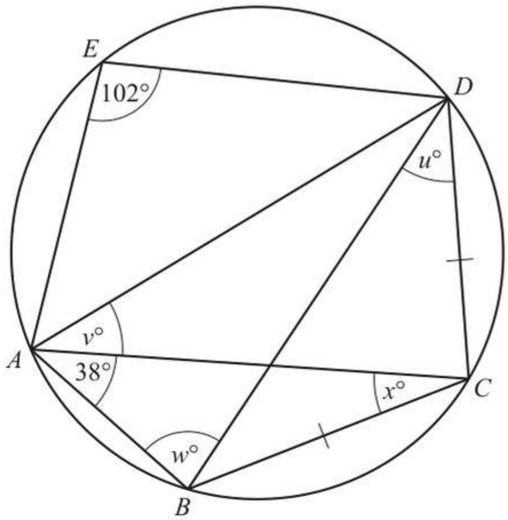
Triangles ABD and ACB are .... [1]

(iii) $AB = 12$ cm, $BC = 11$ cm and $AC = 16$ cm.

Calculate the length of BD.

Answer(a)(iii) BD = ... cm [2]

(b)



NOT TO SCALE

A, B, C, D and E lie on the circle.
Angle $AED = 102°$ and angle $BAC = 38°$.
$BC = CD$.

Find the value of

(i) u,

Answer(b)(i) u = ... [1]

(ii) v,

Answer(b)(ii) v = ... [1]

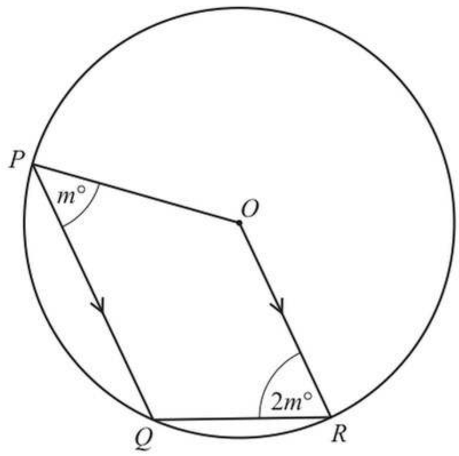
(iii) w,

Answer(b)(iii) w = ... [1]

(iv) x.

Answer(b)(iv) x = ... [1]

(c)



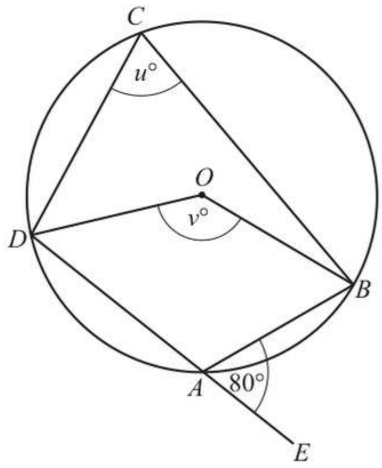
NOT TO SCALE

In the diagram, P, Q and R lie on the circle, centre O.
PQ is parallel to OR.
Angle $QPO = m°$ and angle $QRO = 2m°$.

Find the value of m.

Answer(c) m = ... [5]

8 **(a)**



NOT TO SCALE

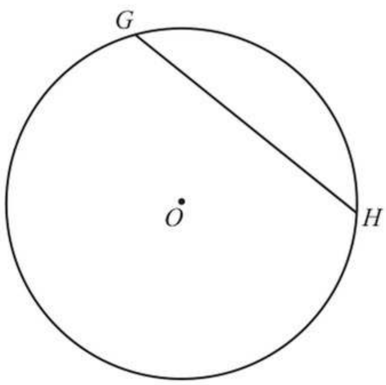
A, B, C and D lie on the circle, centre O.
DAE is a straight line.

Find the value of u and the value of v.

u = ...

v = ... [2]

(b)



NOT TO SCALE

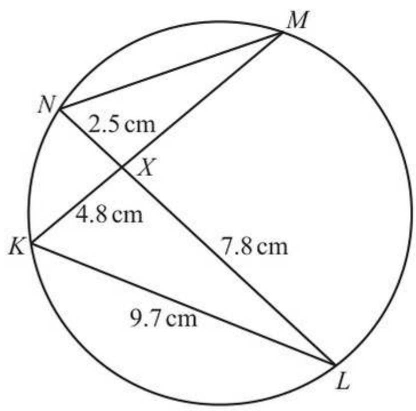
The diagram shows a circle, centre O, radius 8 cm.
GH is a chord of length 10 cm.

Calculate the length of the perpendicular from O to GH.

.... cm [3]

(c) K, L, M and N lie on the circle.
KM and LN intersect at X.
$KL = 9.7$ cm, $KX = 4.8$ cm,
$LX = 7.8$ cm and $NX = 2.5$ cm.

Calculate MN.



$MN =$ .... cm [2]

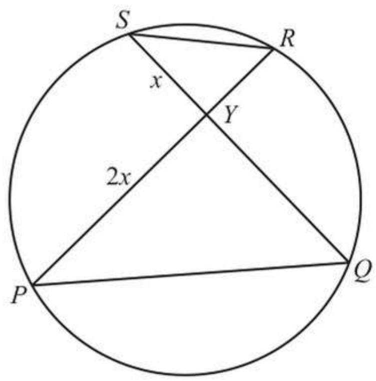
(d) All lengths are in centimetres.

P, Q, R and S lie on the circle.
PR and QS intersect at Y.
$PY = 2x$ and $YS = x$.

The area of triangle $YRS = \frac{5}{12}x(x-1)$.

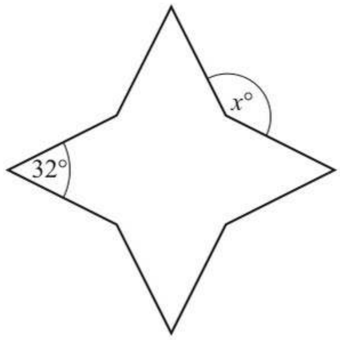
The area of triangle $YQP = x(x+1)$.

Find the value of x.



$x =$ .... [4]

2 **(a)**



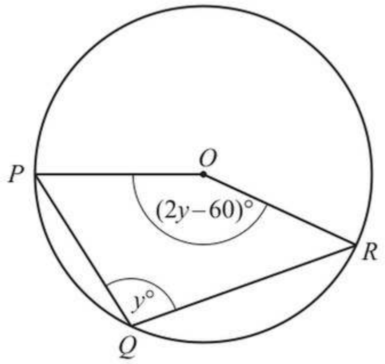
NOT TO SCALE

The diagram shows an octagon.
All of the sides are the same length.
Four of the interior angles are each 32°.
The other four interior angles are equal.

Find the value of x.

$x =$ .... [4]

(b)



NOT TO SCALE

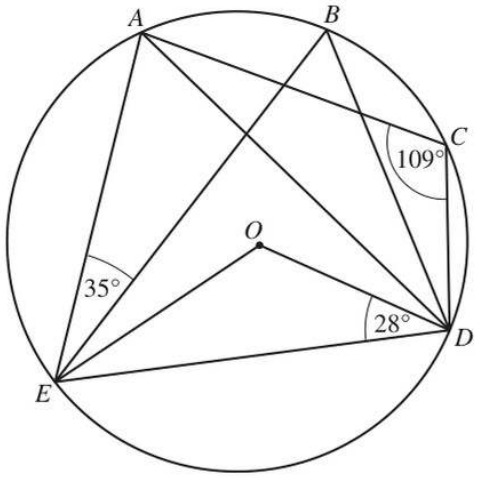
P, Q and R lie on a circle, centre O.
Angle $PQR = y°$ and angle $POR = (2y - 60)°$.

Find the value of y.

$y =$ .... [3]

 0580/41/O/N/17

9 **(a)**



NOT TO SCALE

A, B, C, D and E lie on the circle, centre O.
Angle $AEB = 35°$, angle $ODE = 28°$ and angle $ACD = 109°$.

(i) Work out the following angles, giving reasons for your answers.

(a) Angle EBD = ... because ....

....

.... [3]

(b) Angle EAD = ... because ....

.... [2]

(ii) Work out angle BEO.

Angle BEO = ... [3]

(b) In a regular polygon, the interior angle is 11 times the exterior angle.

(i) Work out the number of sides of this polygon.

.... [3]

(ii) Find the sum of the interior angles of this polygon.

.... [2]

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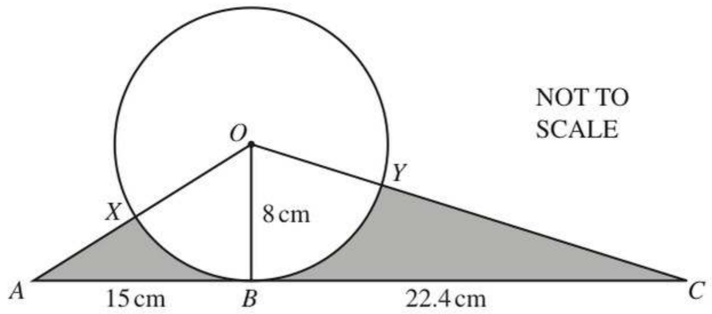
10



The diagram shows a circle, centre O.
The straight line ABC is a tangent to the circle at B.
$OB = 8$ cm, $AB = 15$ cm and $BC = 22.4$ cm.
AO crosses the circle at X and OC crosses the circle at Y.

(a) Calculate angle XOY.

Angle XOY = ... [5]

(b) Calculate the length of the arc XBY.

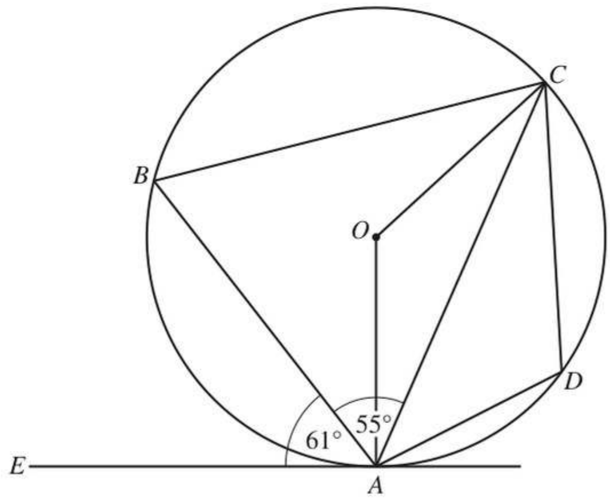
.... cm [2]

(c) Calculate the total area of the two shaded regions.

.... cm^2 [4]

Question 11 is printed on the next page.

7



NOT TO SCALE

In the diagram, A, B, C and D lie on the circle, centre O.
EA is a tangent to the circle at A.
Angle $EAB = 61°$ and angle $BAC = 55°$.

(a) Find angle BAO.

Angle BAO = ... [1]

(b) Find angle AOC.

Angle AOC = ... [2]

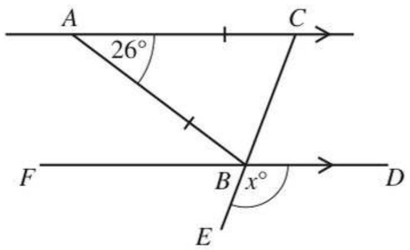
(c) Find angle ABC.

Angle ABC = ... [1]

(d) Find angle CDA.

Angle CDA = ... [1]

2 (a)



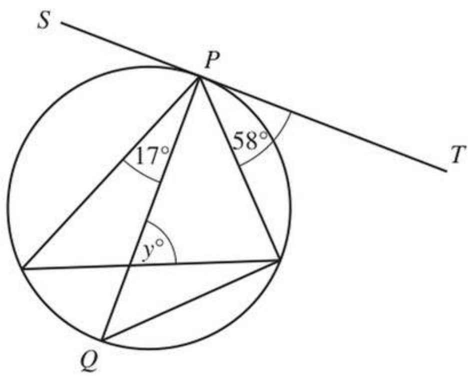
NOT TO SCALE

AC is parallel to *FBD*, *ABC* is an isosceles triangle and *CBE* is a straight line.

Find the value of x.

$x =$ .... [3]

(b)



NOT TO SCALE

The diagram shows a circle with diameter *PQ*.
SPT is a tangent to the circle at *P*.

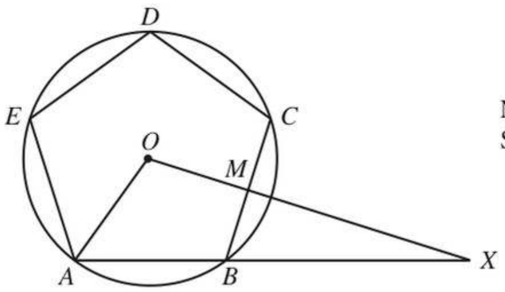
Find the value of y.

$y =$ .... [5]

7 **(a)** Show that each interior angle of a regular pentagon is 108°.

[2]

(b)



The diagram shows a regular pentagon *ABCDE*.
The vertices of the pentagon lie on a circle, centre *O*, radius 12 cm.
M is the midpoint of *BC*.

(i) Find *BM*.

BM = ... cm [3]

(ii) *OMX* and *ABX* are straight lines.

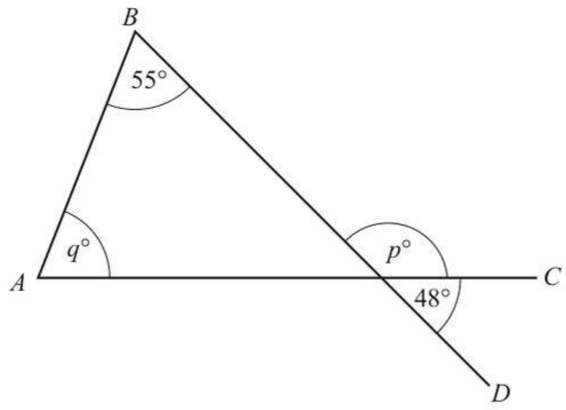
(a) Find *BX*.

BX = ... cm [3]

(b) Calculate the area of triangle *AOX*.

.... cm^2 [3]

1 **(a)**



NOT TO SCALE

In the diagram, AC and BD are straight lines.

Find the value of p and the value of q.

p = ...

q = ... [3]

(b) The angles of a quadrilateral are $x°$, $(x+5)°$, $(2x-25)°$ and $(x+10)°$.

Find the value of x.

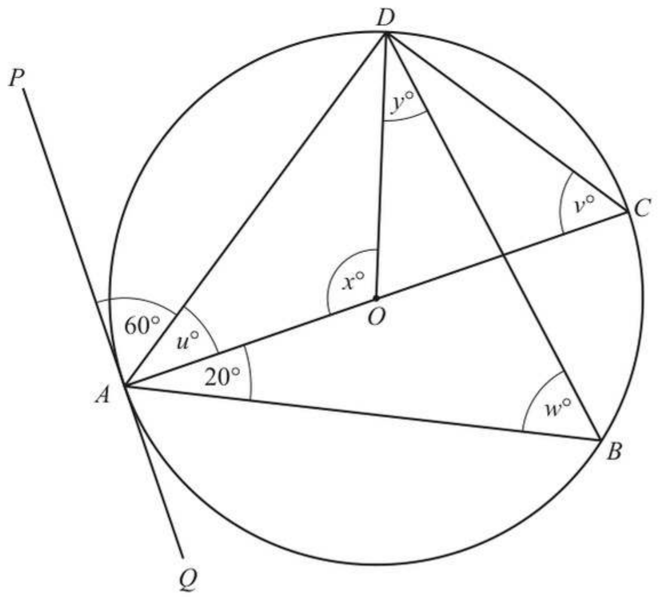
x = ... [3]

(c) A regular polygon has 72 sides.

Find the size of an interior angle.

.... [3]

(d)



NOT TO SCALE

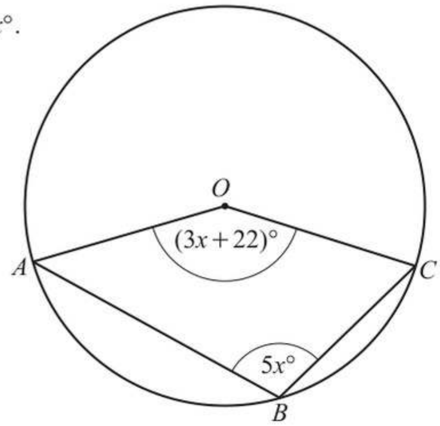
A, B, C and D lie on the circle, centre O, with diameter AC.
PQ is a tangent to the circle at A.
Angle $PAD = 60°$ and angle $BAC = 20°$.

Find the values of u, v, w, x and y.

$u =$ .... , $v =$ .... , $w =$ .... , $x =$ .... , $y =$ .... [6]

(e) A, B and C lie on the circle, centre O.
Angle $AOC = (3x+22)°$ and angle $ABC = 5x°$.

Find the value of x.



NOT TO SCALE

$x =$ .... [4]

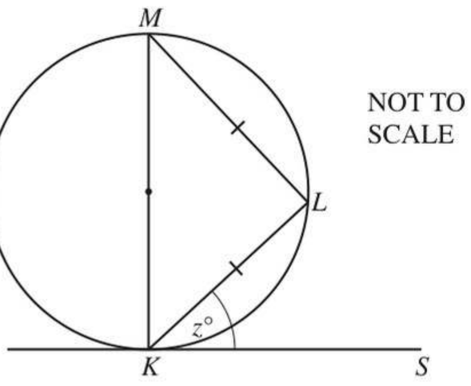
8 **(a)** The interior angle of a regular polygon with n sides is 150°.

Calculate the value of n.

$n =$ .... [2]

(b) **(i)** K, L and M are points on the circle.
KS is a tangent to the circle at K.
KM is a diameter and
triangle KLM is isosceles.

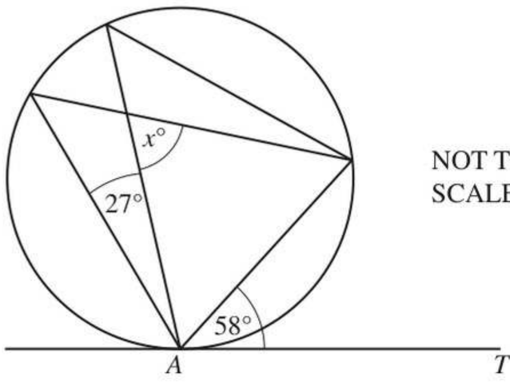
Find the value of z.



$z =$ .... [2]

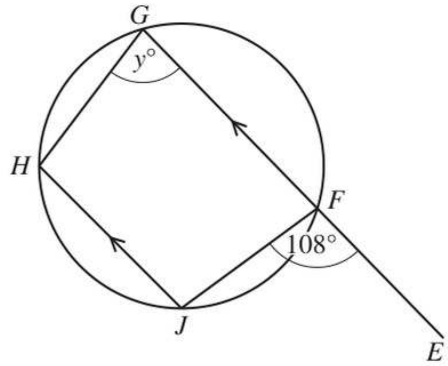
(ii) AT is a tangent to the circle at A.

Find the value of x.



$x =$ .... [2]

(iii)



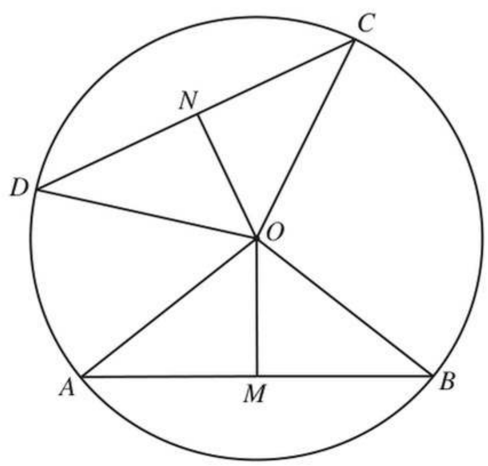
NOT TO SCALE

F, G, H and J are points on the circle.
EFG is a straight line parallel to JH.

Find the value of y.

$y =$ .... [2]

(c)



NOT TO SCALE

A, B, C and D are points on the circle, centre O.
M is the midpoint of AB and N is the midpoint of CD.
$OM = ON$

Explain, giving reasons, why triangle OAB is congruent to triangle OCD.

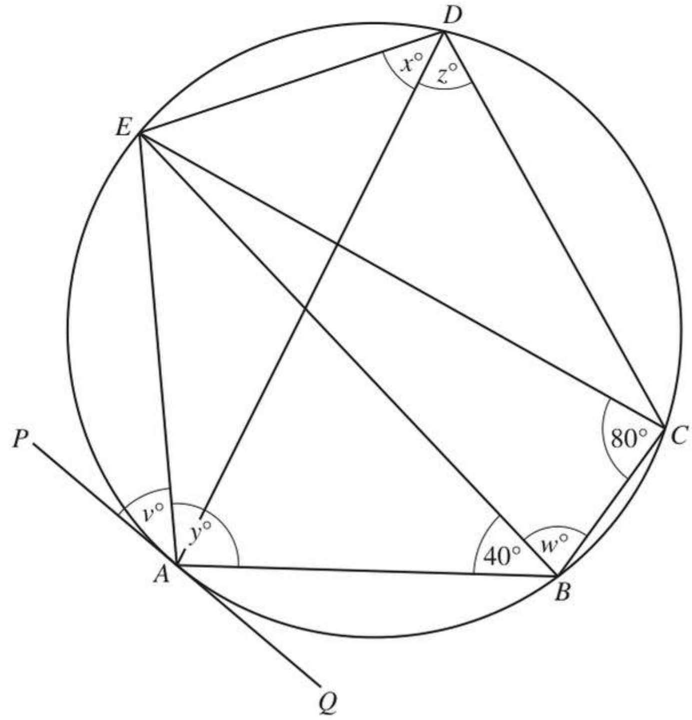
....

....

....

.... [3]

8 **(a)**

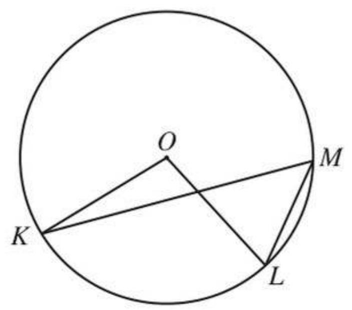


The points A, B, C, D and E lie on the circle.
PAQ is a tangent to the circle at A and $EC = EB$.
Angle $ECB = 80°$ and angle $ABE = 40°$.

Find the values of v, w, x, y and z.

v = ... w = ... x = ... y = ... z = ... [5]

(b)

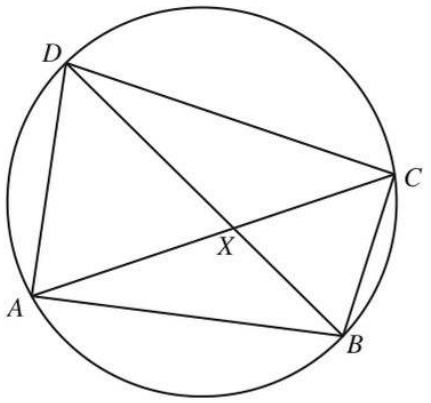


In the diagram, K, L and M lie on the circle, centre O.
Angle $KML = 2x°$ and reflex angle $KOL = 11x°$.

Find the value of x.

x = ... [3]

(c)



The diagonals of the cyclic quadrilateral $ABCD$ intersect at X.

(i) Explain why triangle ADX is similar to triangle BCX.
Give a reason for each statement you make.

....

....

....

.... [3]

(ii) $AD = 10$ cm, $BC = 8$ cm, $BX = 5$ cm and $CX = 7$ cm.

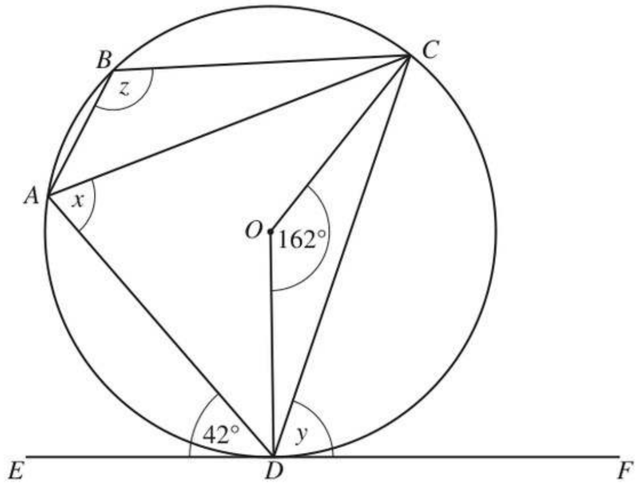
(a) Calculate DX.

$DX =$ .... cm [2]

(b) Calculate angle BXC.

Angle $BXC =$ .... [4]

5 **(a)**



NOT TO SCALE

A, B, C and D are points on the circle, centre O.
EF is a tangent to the circle at D.
Angle $ADE = 42°$ and angle $COD = 162°$.

Find the following angles, giving reasons for each of your answers.

(i) Angle x

$x =$ .... because ....

.... [2]

(ii) Angle y

$y =$ .... because ....

.... [2]

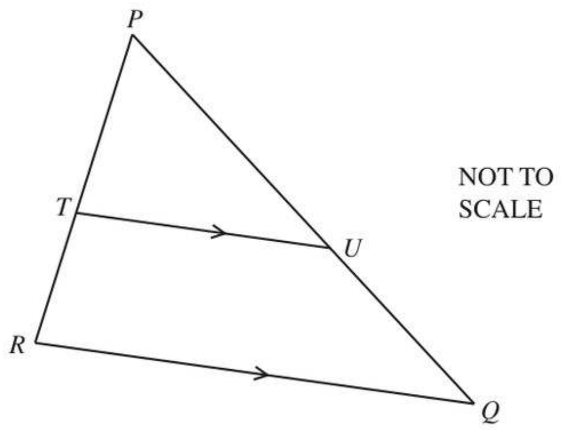
(iii) Angle z

$z =$ .... because ....

....

.... [3]

(b)



PQR is a triangle.
T is a point on *PR* and *U* is a point on *PQ*.
RQ is parallel to *TU*.

(i) Explain why triangle *PQR* is similar to triangle *PUT*.
Give a reason for each statement you make.

....

....

....

.... [3]

(ii) $PT : TR = 4 : 3$

(a) Find the ratio $PU : PQ$.

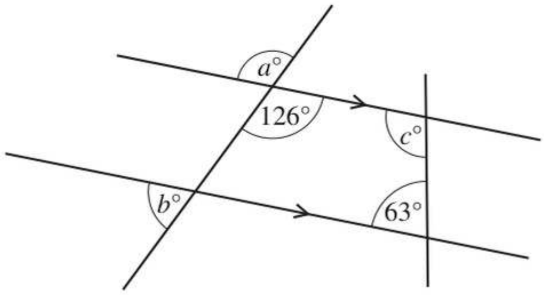
.... : .... [1]

(b) The area of triangle *PUT* is 20 cm^2.

Find the area of the quadrilateral *QRTU*.

.... cm^2 [3]

3 **(a)**



NOT TO SCALE

The diagram shows two straight lines intersecting two parallel lines.

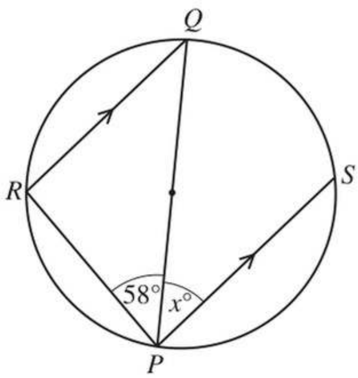
Find the values of a, b and c.

a = ...

b = ...

c = ... [3]

(b)



NOT TO SCALE

Points R and S lie on a circle with diameter PQ.
RQ is parallel to PS.
Angle $RPQ = 58°$.

Find the value of x, giving a geometrical reason for each stage of your working.

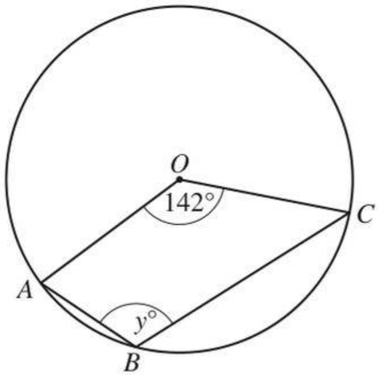
....

....

....

x = ... [3]

(c)



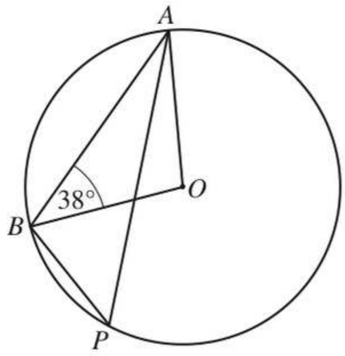
NOT TO SCALE

Points A, B and C lie on a circle, centre O.
Angle $AOC = 142°$.

Find the value of y.

$y =$ .... [2]

2 **(a)**



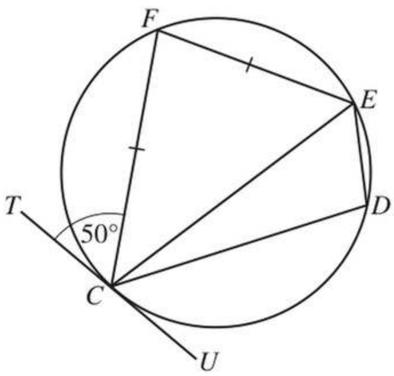
NOT TO SCALE

A, B and P are points on a circle, centre O and angle $OBA = 38°$.

Find angle APB.

Angle APB = ... [3]

(b)



NOT TO SCALE

$CDEF$ is a cyclic quadrilateral and $FC = FE$.
TU is a tangent to the circle at C and angle $TCF = 50°$.

Find

(i) angle EFC,

Angle EFC = ... [2]

(ii) angle CDE.

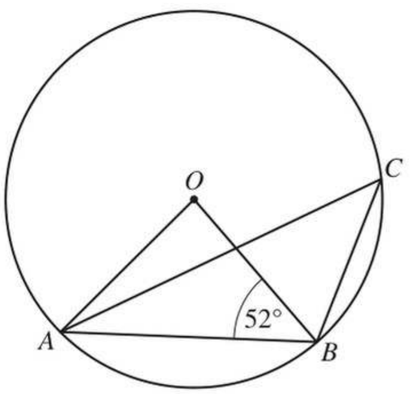
Angle CDE = ... [1]

6 **(a)** The interior angle of a regular polygon is 156°.

Calculate the number of sides of this polygon.

.... [2]

(b)



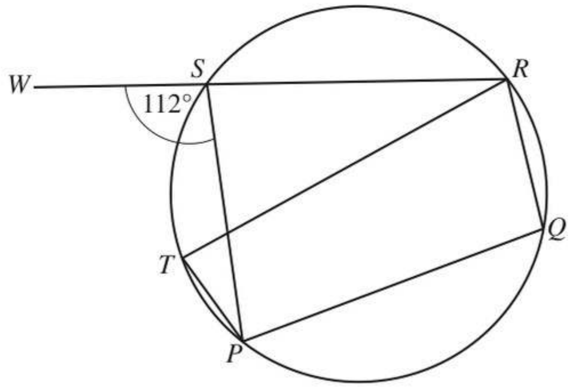
NOT TO SCALE

A, B and C lie on a circle, centre O.
Angle $OBA = 52°$.

Calculate angle ACB.

Angle $ACB =$ .... [2]

(c)

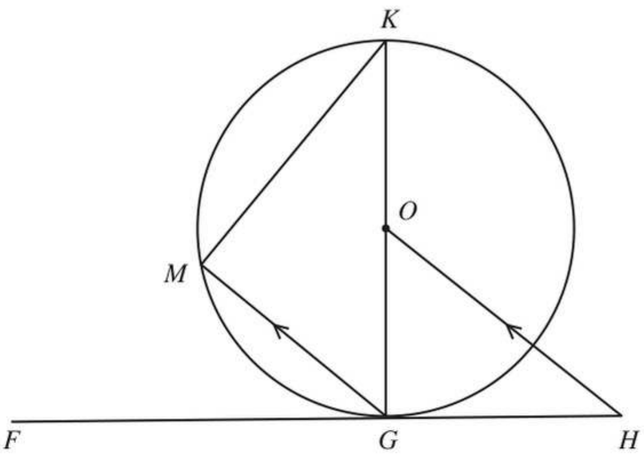


P, *Q*, *R*, *S* and *T* lie on a circle.
WSR is a straight line and angle *WSP* = 112°.

Calculate angle *PTR*.

Angle *PTR* = ... [2]

(d)



G, *K* and *M* lie on a circle, centre *O*.
FGH is a tangent to the circle at *G* and *MG* is parallel to *OH*.

Show that triangle *GKM* is mathematically similar to triangle *OHG*.
Give a geometrical reason for each statement you make.

....

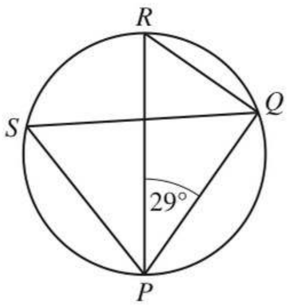
....

....

....

.... [4]

2 **(a)**



NOT TO SCALE

The points P, Q, R and S lie on a circle with diameter PR.

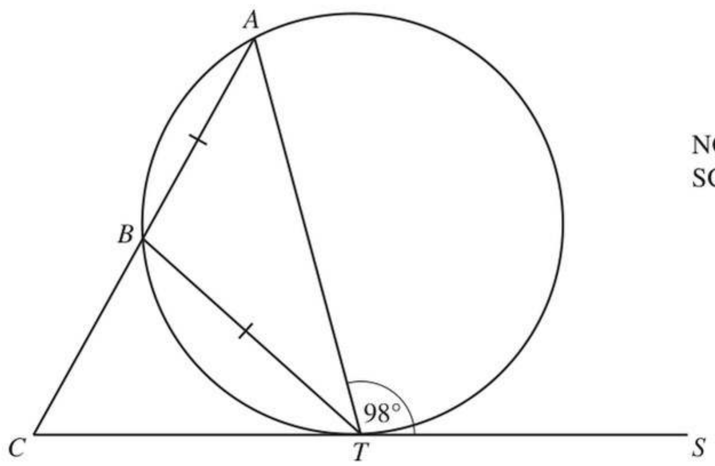
Work out the size of angle PSQ, giving a geometrical reason for each step of your working.

....

....

.... [3]

(b)



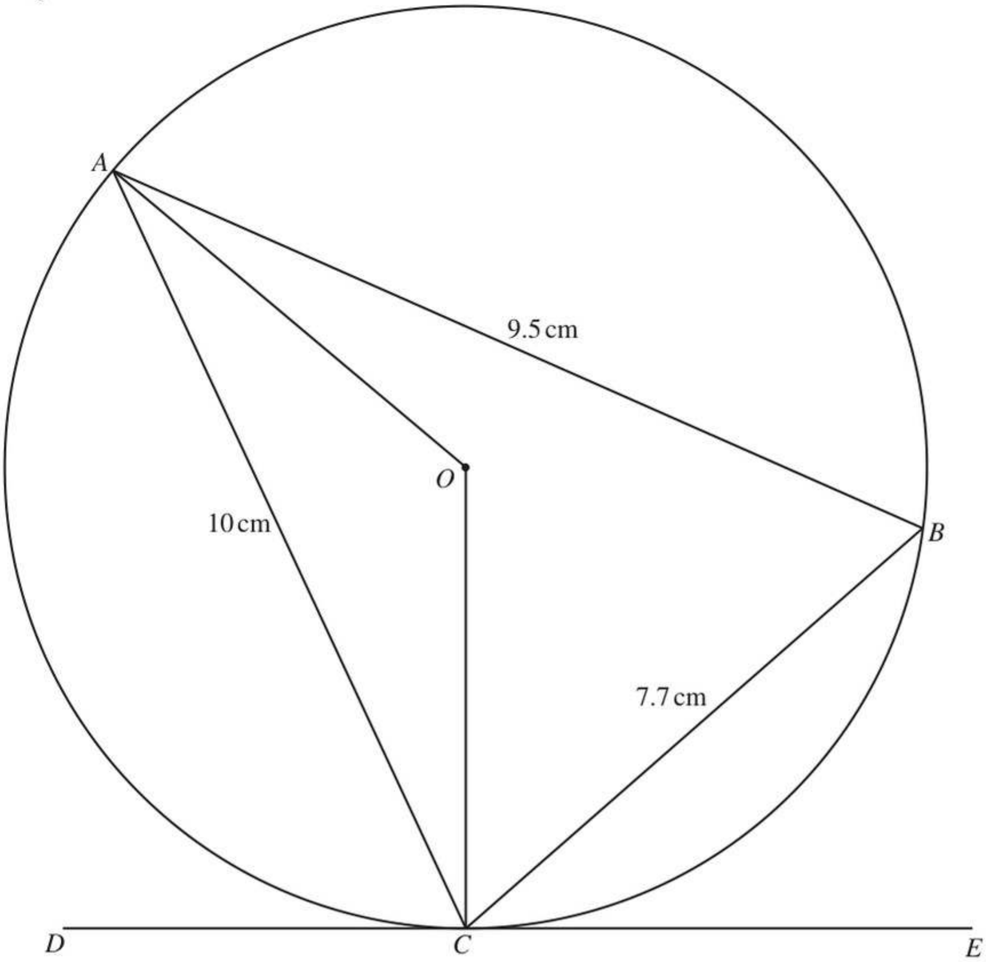
NOT TO SCALE

The points A, B and T lie on a circle and CTS is a tangent to the circle at T.
ABC is a straight line and $AB = BT$.
Angle $ATS = 98°$.

Work out the size of angle ACT.

Angle ACT = ... [4]

8



NOT TO SCALE

A, B and C are points on the circle, centre O.
DE is a tangent to the circle at C.
$AC = 10\,\text{cm}$, $AB = 9.5\,\text{cm}$ and $BC = 7.7\,\text{cm}$.

(a) Show that angle $ABC = 70.2°$, correct to 1 decimal place.

[4]

(b) Find

(i) angle AOC

Angle AOC = ... [1]

(ii) angle ACO

Angle ACO = ... [1]

(iii) angle ACD.

Angle ACD = ... [1]

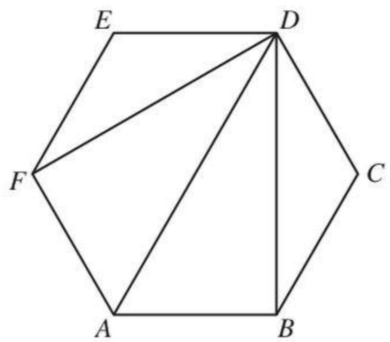
(c) Calculate the radius, OC, of the circle.

OC = ... cm [3]

(d) Calculate the area of triangle ABC as a percentage of the area of the circle.

.... % [4]

10 (a)



NOT TO SCALE

ABCDEF is a regular hexagon.
DF, *DA* and *DB* are diagonals.

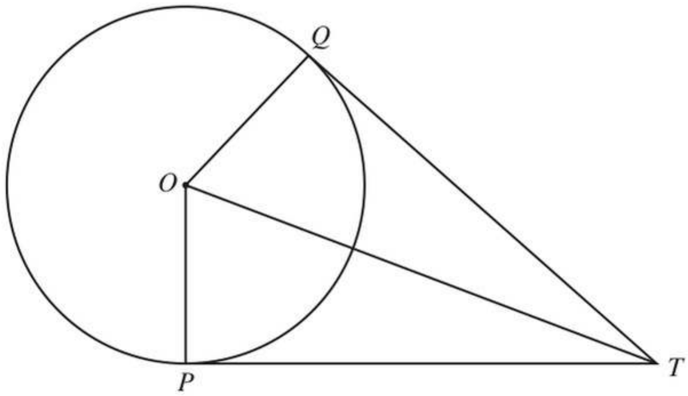
Complete the following statements using three different triangles.

Triangle *DEF* is congruent to triangle ....

Triangle .... is congruent to triangle ....

[2]

(b)



NOT TO SCALE

P and *Q* are points on the circle with centre *O*.
TP and *TQ* are tangents to the circle from the point *T*.

Complete the following statements and reasons.

In triangles *OPT* and *OQT*

OP = ... because each is a radius of the circle

OT is a common side

Angle *OPT* = angle .... = 90° because ....

Triangles *OPT* and *OQT* are congruent using the criterion ....

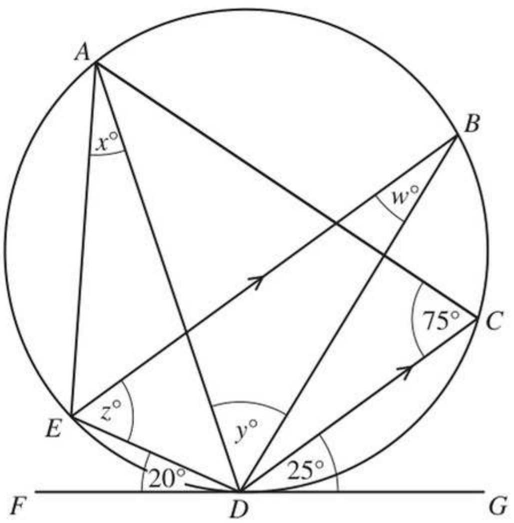
This proves that the tangents *TP* and *TQ* are ....

[5]

4 **(a)** Find the size of one interior angle of a regular 10-sided polygon.

.... [2]

(b)



NOT TO SCALE

The points A, B, C, D and E lie on a circle.
FG is a tangent to the circle at D.
EB is parallel to DC.

Find the value of each of w, x, y and z.

w = ...

x = ...

y = ...

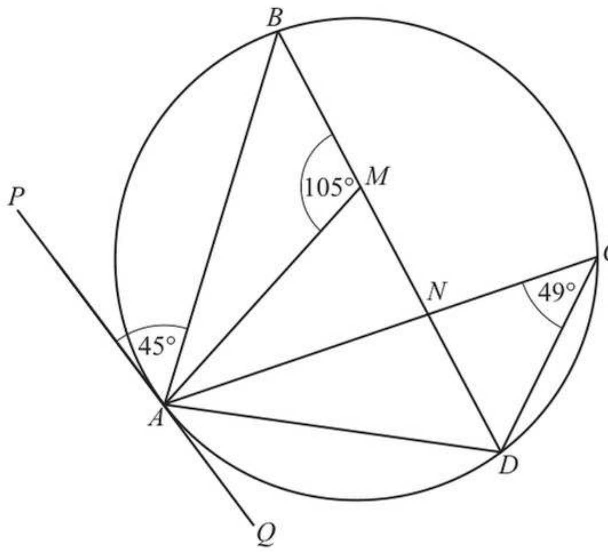
z = ... [5]

4 **(a)** The angles of a quadrilateral are $w°$, $x°$, $y°$ and $z°$.
The ratio $w : (x+y+z) = 3 : 5$.

Find the value of w.

$w =$ .... [2]

(b)



NOT TO SCALE

A, B, C and D are points on a circle.
PQ is the tangent to the circle at A.
$BMND$ is a straight line.
Angle $ACD = 49°$, angle $AMB = 105°$ and angle $PAB = 45°$.

(i) Find angle BAM.

Angle $BAM =$ .... [2]

(ii) **(a)** Find angle BAD.

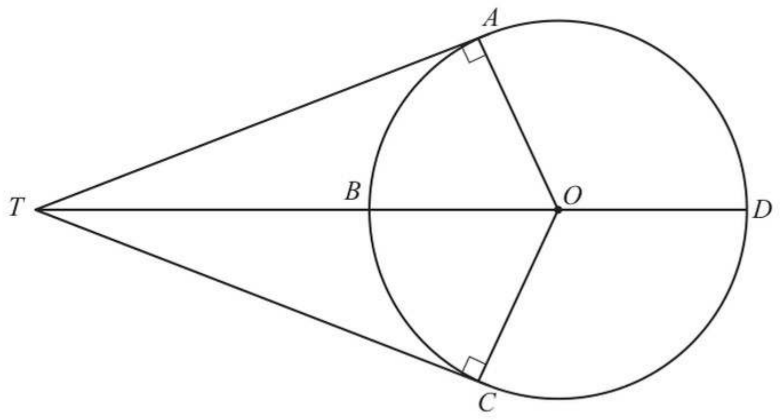
Angle $BAD =$ .... [2]

(b) Give a geometrical reason why BD is **not** the diameter of the circle.

....

.... [1]

(c)



NOT TO SCALE

A, B, C and D are points on a circle, centre O.
TA and TC are tangents to the circle.
$OA = 6.75$ cm and $OT = 11.5$ cm.

(i) Show that angle $AOC = 108.12°$, correct to 2 decimal places.

[3]

(ii) Calculate the length of the **minor** arc ABC.

.... cm [2]

(iii) Calculate the area of the **major** sector $OCDA$.

.... cm^2 [3]

0580 | PAPER 4

SECTION 04.2

Constructions, loci and scale drawings

Sorted by session, paper variant and original question number

4
QUESTIONS

6
EXAM PAGES

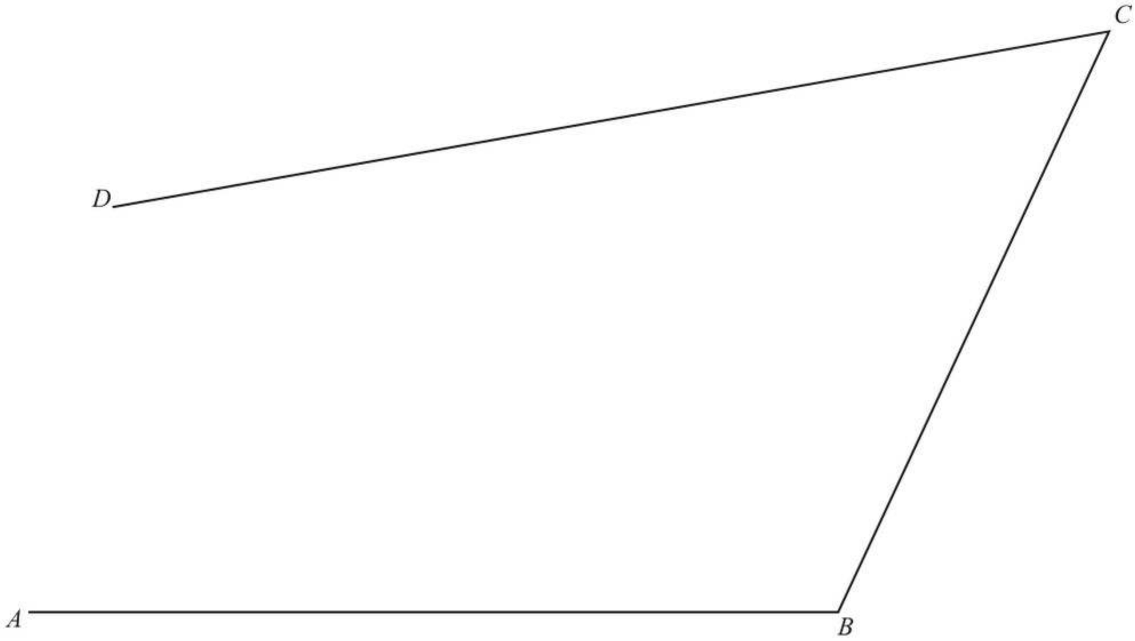
P4
CALCULATOR

TRACK YOUR ATTEMPTS | WRITE FULL WORKING | CHECK ACCURACY | REVIEW ERRORS

04.2 Constructions, loci and scale drawings

#	SOURCE QUESTION	PAGES	DONE	MARK	REVIEW
01	2015 May June Paper 41 | Question 10	2	☐	___	☐
02	2016 Feb March Paper 42 | Question 2	1	☐	___	☐
03	2018 May June Paper 41 | Question 2	1	☐	___	☐
04	2019 Feb March Paper 42 | Question 4	2	☐	___	☐

10 The diagram is a scale drawing of three straight roads, *AB*, *BC* and *CD*.
The scale is 1 : 5000.



Scale 1 : 5000

(a) Find the actual length of the road *BC*.
Give your answer in metres.

Answer(a) .... m [2]

(b) Another straight road starts at *M*, the midpoint of *AB*.
This road is perpendicular to *AB* and it meets the road *CD* at *X*.

Using a straight edge and compasses only, construct *MX*. [2]

(c) There is a park in the area enclosed by the four roads.

The park is

- less than 290 m from *B*

and

- nearer to *CD* than to *CB*.

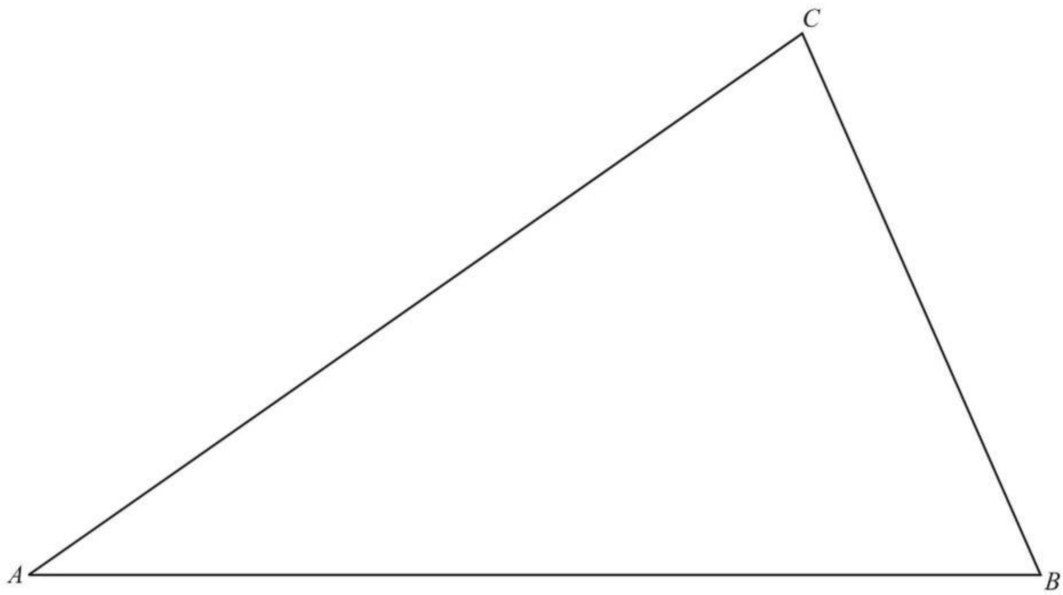
Using a ruler and compasses only, construct the boundaries of the park.

Leave in all your construction arcs and label the park *P*. [5]

Question 11 is printed on the next page.

2 In this question use a ruler and compasses only.
Show all your construction arcs.

The diagram shows a triangular field *ABC*.
The scale is 1 centimetre represents 50 metres.



Scale : 1 cm to 50 m

(a) Construct the locus of points that are equidistant from *A* and *B*. [2]

(b) Construct the locus of points that are equidistant from the lines *AB* and *AC*. [2]

(c) The two loci intersect at the point *E*.

Construct the locus of points that are 250 m from *E*. [2]

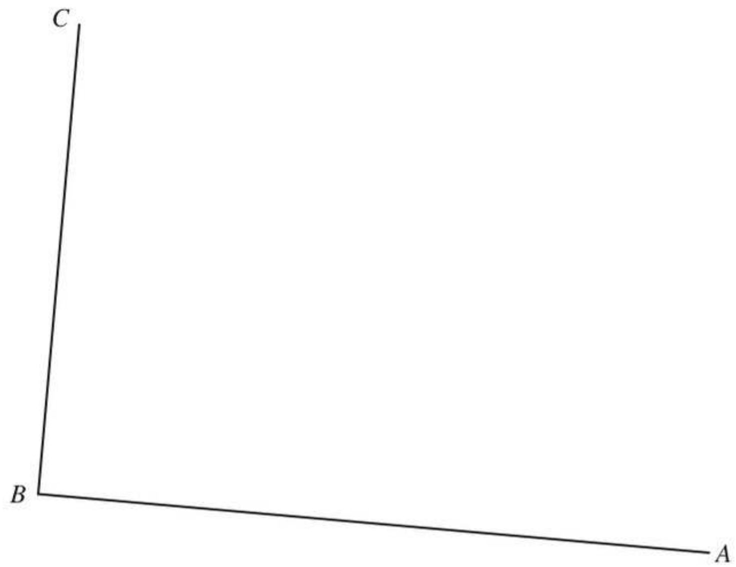
(d) Shade any region inside the field *ABC* that is

- more than 250 m from *E*

and

- closer to *AC* than to *AB*. [2]

2 The scale drawing shows two boundaries, AB and BC, of a field $ABCD$.
The scale of the drawing is 1 cm represents 8 m.



Scale: 1 cm to 8 m

(a) The boundaries CD and AD of the field are each 72 m long.

(i) Work out the length of CD and AD on the scale drawing.

.... cm [1]

(ii) **Using a ruler and compasses only**, complete accurately the scale drawing of the field. [2]

(b) A tree in the field is

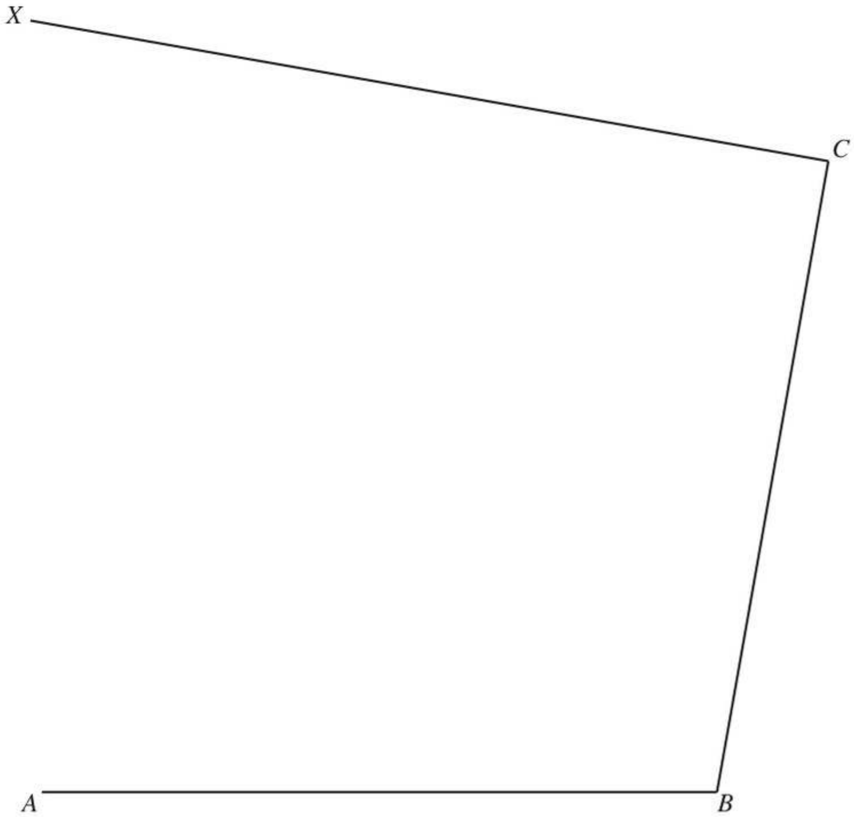
- equidistant from A and B

and

- equidistant from AB and BC.

On the scale drawing, construct two lines to find the position of the tree.
Use a straight edge and compasses only and leave in your construction arcs. [4]

4 The diagram shows an incomplete scale drawing of a market place, *ABCD*, where *D* is on *CX*.
The scale is 1 centimetre represents 5 metres.



Scale : 1 cm to 5 m

D lies on *CX* such that angle $DAB = 75°$.

(a) On the diagram, draw the line *AD* and mark the position of *D*. [2]

(b) Find the actual length of the side *BC* of the market place.

.... m [2]

(c) **In this part, use a ruler and compasses only.**

Street sellers are allowed in the part of the market place that is

- more than 35 metres from *A*

and

- nearer to *C* than to *B*

and

- nearer to *CD* than to *BC*.

On the diagram, construct and shade the region where street sellers are allowed. [7]

(d) Write the scale of the drawing in the form $1 : n$.

1 : .... [1]

0580 | PAPER 4

SECTION 04.3

Similarity and congruence

Sorted by session, paper variant and original question number

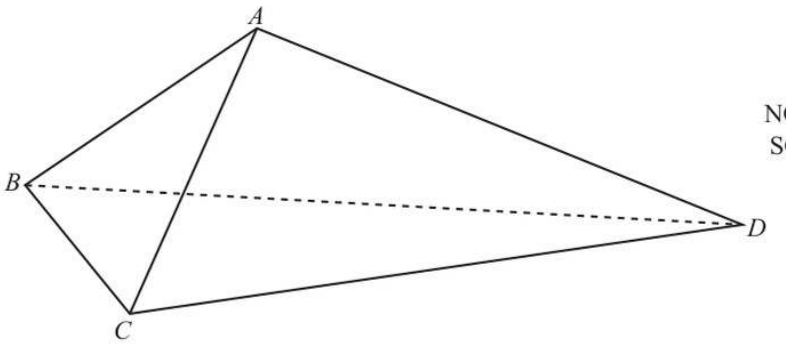
2	3	P4
QUESTIONS	EXAM PAGES	CALCULATOR

TRACK YOUR ATTEMPTS | WRITE FULL WORKING | CHECK ACCURACY | REVIEW ERRORS

04.3 Similarity and congruence

#	SOURCE QUESTION	PAGES	DONE	MARK	REVIEW
01	2015 Oct Nov Paper 42 | Question 4	2	☐	___	☐
02	2022 May June Paper 42 | Question 9	1	☐	___	☐

4



The diagram shows a tent *ABCD*.
The front of the tent is an isosceles triangle *ABC*, with $AB = AC$.
The sides of the tent are congruent triangles *ABD* and *ACD*.

(a) $BC = 1.2$ m and angle $ABC = 68°$.

Find *AC*.

Answer(a) $AC =$ .... m [3]

(b) $CD = 2.3$ m and $AD = 1.9$ m.

Find angle *ADC*.

Answer(b) Angle $ADC =$ .... [4]

(c) The floor of the tent, triangle *BCD*, is also an isosceles triangle with $BD = CD$.

Calculate the area of the floor of the tent.

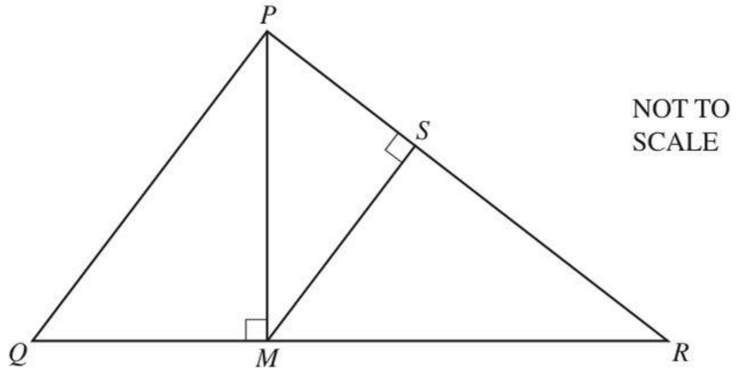
Answer(c) .... m^2 [4]

(d) When the tent is on horizontal ground, *A* is a vertical distance 1.25 m above the ground.

Calculate the angle between *AD* and the ground.

Answer(d) .... [3]

9 **(a)**



In triangle PQR, M lies on QR and S lies on PR.

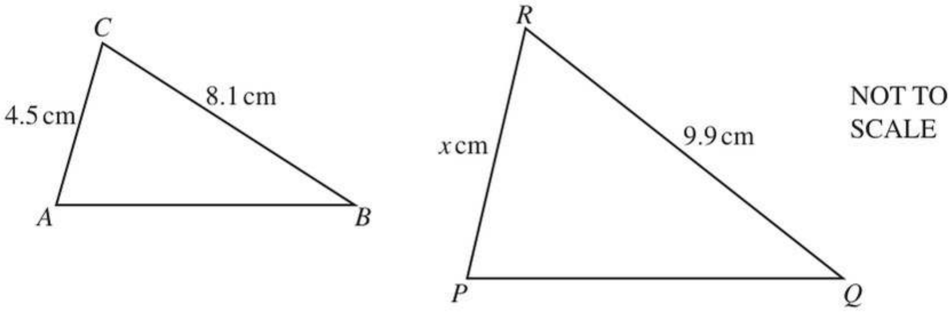
Explain, giving reasons, why triangle PMR is similar to triangle MSR.

....

....

.... [3]

(b)



Triangle ABC is similar to triangle PQR.

(i) Find the value of x.

x = ... [2]

(ii) The area of triangle PQR is 25 cm^2.

Calculate the area of triangle ABC.

.... cm^2 [2]

